

RESEARCH INTERESTS

Tackling ill-posed problems across **3D Vision & Robotics**

Recent research focuses: (i) Vision foundation model for Low-light/Scattering conditions (ii) Tactile prediction from Egocentric video

EDUCATION

University of Minnesota

M.S. in Robotics

MN, United States

Sep. 2025 — Present

Pusan National University

B.S. in Mechanical Engineering

South Korea

Mar. 2019 — Feb. 2025

2 years of absence due to obligatory military service (2020 — 2022)

PUBLICATIONS

8. **DarkVGGT: Seeing Through Darkness Using Thermal Geometry without Daylight Tax**

Minseong Kweon, Wenyuan Zhao, Nuo Chen, Lulin Liu, Huiwen Han, Zihao Zhu, Srinivas Shakkottai, Chao Tian, Zhiwen Fan
Under Review (NeurIPS 2026) [\[Paper\]](#) [\[Project Page\]](#)

7. **Swimm3R: Splatting with Medium-aware SfM for Underwater 3D Reconstruction**

Minseong Kweon, Junaed Sattar
Under Review (IEEE RA-L) [\[Paper\]](#) [\[Project Page\]](#)

6. **OceanSplat: Object-aware Gaussian Splatting with Trinocular View Consistency for Underwater Scene Reconstruction**

Minseong Kweon, Jinsun Park
AAAI 2026 [\[Paper\]](#) [\[Project Page\]](#)

5. **All-day Depth Completion via Thermal-LiDAR Fusion**

Janghyun Kim, **Minseong Kweon**, Jinsun Park, Ukcheol Shin
Under Review (IEEE T-IV) [\[Paper\]](#)

4. **CAR-Stereo: Confidence-aware Adaptive Disparity Refinement for Real-time Stereo Matching**

Chanill Park, Janghyun Kim, **Minseong Kweon**, Jinsun Park
IEEE RA-L 2025 [\[Paper\]](#)

3. **MrGS: Multi-modal Radiance Fields with 3D Gaussian Splatting for RGB-Thermal Novel View Synthesis**

Minseong Kweon, Janghyun Kim, Ukcheol Shin, Jinsun Park
ICRA 2025 Thermal Infrared in Robotics Workshop [\[Paper\]](#)
* Received **Best Poster Award**

2. **NoiseGS: Boosting 3D Gaussian Splatting with Positional Noise for Large-Scale Scene Reconstruction**

Minseong Kweon, Kai Cheng, Xuejin Chen, Jinsun Park
Eurographics 2025 Short paper [\[Paper\]](#)

1. **ULTRON: Unifying Local Transformer and Convolution for Large-Scale Image Retrieval**

Minseong Kweon, Jinsun Park
ACCV 2024 [\[Paper\]](#)

EXPERIENCE

GRASP Lab, University of Pennsylvania

Research Intern (Advisor: Prof. Lingjie Liu)

PA, United States

May 2026 — Present

- Conducted research on tactile prediction from egocentric video for dexterous robotic manipulation. [\[Preview Result\]](#)

PHAI Lab, Texas A&M University

Research Intern (Advisor: Prof. Zhiwen Fan)

(remote) TX, United States

Jan. 2026 — May 2026

- Conducted research on feed-forward 3D reconstruction model using thermal geometry in low-light scenes. → **DarkVGGT**

- IRV Lab, University of Minnesota

Research Intern (Advisor: Prof. Junaed Sattar)

MN, United States

Jan. 2026 — May 2026

• Conducted research on a feed-forward 3D reconstruction model for highly turbid underwater videos. → Swimm3R
- NEL-BITA, University of Science and Technology of China

Research Intern (Advisor: Prof. Xuejin Chen)

China

Jul. 2024 — Aug. 2024

• Conducted research on large-scale 3D reconstruction using multi-GPU 3D Gaussian Splatting with noise injection. → NoiseGS
- Artificial Intelligence Creative Research Laboratory, ETRI

Research Intern (Advisor: Dr. Seung-Min Choi)

South Korea

Jul. 2023 — Aug. 2023

• Conducted research on instance-level feature representations for large-scale landmark image retrieval. → ULTRON

AWARDS & HONORS

6. Albert Howard Award , Professional Development Funding

Issued by University of Minnesota

May 2026
5. Best Poster Award, Thermal Infrared in Robotics Workshop, IEEE ICRA 2025

Issued by IEEE Robotics and Automation Society

May 2025
4. Honorable Mention, Undergraduate AI Paper Competition, Summer Annual Conference of IEIE 2024

Issued by Institute of Electronics and Information Engineers

Jun. 2024
3. Association Presidential Award, World Creative Robot Contest

Issued by Robot Education Contents Association

Oct. 2023
2. Excellence Award, DAOL Venture Challenge

Issued by DAOL Financial Group

Nov. 2022
1. Presidential Award, PNU IT Idea Arena

Issued by Pusan National University

Oct. 2022

PRESENTATIONS

3. Self-Similarity Feature Learning for Person Re-Identification

Minseong Kweon, Jinsun Park

Summer Annual Conference of the Institute of Electronics and Information Engineers (IEIE), 2024 [\[Paper \(in Korean\)\]](#)

* Received Honorable Mention
2. Hierarchical Visual Localization for Indoor Autonomous Systems

Minseong Kweon, Jeonghyun Noh, Chanill Park, Jinsun Park

Korea Robotics Conference (KRoC), 2024 [\[Paper \(in Korean\)\]](#)
1. Attention-based Human Activity Recognition with 3-axis Accelerometer Data Conversion

Minseong Kweon, Jaehyeong Park, Kyunghyun Kim, Jeonghyun Noh, Jinsun Park

Korea Computer Congress (KCC), 2023 [\[Paper \(in Korean\)\]](#)

ACADEMIC SERVICES

2. Program Committee & Reviewer AAAI Conference on Artificial Intelligence (AAAI), 2026, 2027
1. Program Committee & Reviewer Asian Conference on Computer Vision (ACCV), 2026

PATENTS

1. Minseong Kweon, Jinsun Park, “METHOD AND SYSTEM FOR PROCESSING IMAGES BASED ON LOCAL ATTENTION”.

Korean Patent Application No. 10-2024-0176234. Patent Pending.

SKILLS

Languages	C/C++, Python, MATLAB, JavaScript
Tools	Github, Linux, ROS, Wandb, CUDA programming
Libraries	NumPy, SciPy, Pytorch, Pytorch3D, OpenCV, Open3D, PyBullet, PyRender, scikit-learn